Module 1 — Practical Skills in Physics

This module underpins all experimental work in A-Level Physics and is assessed across all papers, including the Practical Endorsement (PAG).

1.1 Planning Experiments

Variables

Independent Variable (IV): The variable you deliberately change in the experiment.
Dependent Variable (DV): The variable you measure as a result of changing the IV.
Control Variables: All other variables that must be kept constant to ensure a fair test.

When planning an experiment you must:

- Clearly state the IV, DV, and all significant control variables
- Describe how each control variable will be kept constant
- Specify the range and intervals of the IV (at least 5-6 values, spread across a suitable range)
- Describe the method for measuring the DV, including the instrument and its resolution
- State how many repeat readings will be taken (typically at least 3)
- Identify potential safety hazards and precautions

Exam Tip: When asked to "describe an experiment," always state what you measure, what you change, what you keep constant, and how you will process the data (e.g., plot a graph of DV vs IV and find the gradient).

Apparatus Selection

Choose instruments appropriate to the quantity being measured:

Quantity	Instrument	Typical Resolution
Length (small)	Micrometer screw gauge	0.01 mm
Length (small)	Vernier callipers	0.1 mm or 0.02 mm
Length (medium)	Ruler	1 mm
Mass	Digital balance	0.01 g or 0.1 g
Time	Stopwatch	0.01 s
Temperature	Thermometer / thermocouple	0.5 °C / 0.1 °C
Voltage	Voltmeter / multimeter	0.01 V
Current	Ammeter / multimeter	0.01 A

1.2 Data Collection

Recording Data in Tables

- Draw a ruled table before starting the experiment
- The IV goes in the first column; the DV and calculated quantities in subsequent columns
- Every column heading must include the quantity name and its unit, e.g., `Length / m` or `Length (m)`
- All raw data in a column should be recorded to the same number of decimal places (matching the resolution of the instrument)
- Include columns for repeat readings and a mean

Significant Figures

> **Significant figures (s.f.):** The number of meaningful digits in a measurement. Leading zeros are not significant; trailing zeros after a decimal point are significant.

Rules for significant figures in calculations:

- **Multiplication / Division:** The answer should have the same number of s.f. as the value with the fewest s.f.
- **Addition / Subtraction:** The answer should have the same number of decimal places as the value with the fewest d.p.
- Final answers are typically given to 2 or 3 s.f. in A-Level Physics
- Do not round intermediate calculations — only round the final answer

Exam Tip: If the question gives data to 3 s.f., give your answer to 2 or 3 s.f. Never give more significant figures in your answer than in the data you were given.

1.3 Graphs

Plotting Graphs

- The IV (or the variable you chose values for) goes on the x-axis; the DV goes on the y-axis
- Both axes must be labelled with the quantity and unit
- Choose scales so that the plotted points fill at least half of the graph paper in each direction
- Use simple, easy-to-read scales (e.g., multiples of 1, 2, 5, 10)
- Plot points with small, precise crosses (not large dots or circles)
- The origin (0, 0) does not need to be included unless it is a data point or is physically meaningful

Lines of Best Fit

- A **straight line of best fit** should pass through or close to as many points as possible, with roughly equal numbers of points above and below the line
- A **curve of best fit** should be a smooth curve that follows the trend of the points — do not join point-to-point
- Anomalous points should be circled and excluded from the line of best fit

Gradients

Gradient = rise / run = $\Delta y / \Delta x$

- Use a large triangle (at least half the length of the line) for accuracy
- Read values from the line of best fit, not from data points
- Include the unit of the gradient (unit of y-axis / unit of x-axis)
- For a curve, the gradient at a point is found by drawing a tangent to the curve at that point, then calculating the gradient of the tangent

Y-Intercept

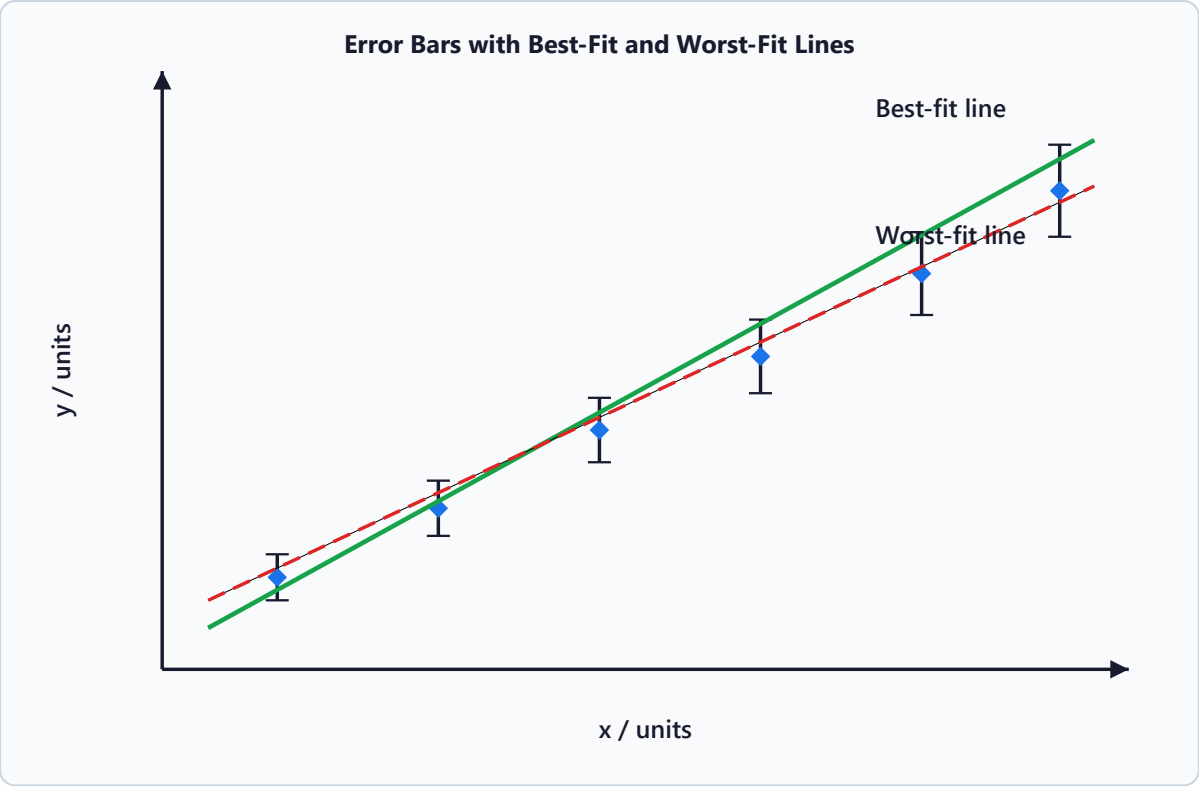
- Read directly from the graph where the line of best fit crosses the y-axis
- If the y-axis does not start at zero, calculate the intercept using: `c = y - mx` using a point on the line

Interpreting Graphs

Graph Shape	Relationship	Equation Form
Straight line through origin	Directly proportional	$y = kx$
Straight line with intercept	Linear	$y = mx + c$
Curve ($y = kx^2$)	Plot y vs x^2 to linearise	$y = kx^2$
Curve ($y = k/x$)	Plot y vs 1/x to linearise	$y = k/x$
Exponential	Plot ln y vs x	$y = Ae^{kx}$

Exam Tip: To show two quantities are proportional, plot them against each other and check for a straight line through the origin. If it is linear but not through the origin, they are linearly related but not proportional.

Error Bars



- Error bars show the uncertainty in each data point
- They are drawn as vertical (and sometimes horizontal) lines extending above and below (or either side of) each data point
- The length of each error bar equals the uncertainty in that measurement (or half the range of repeats)
- A worst line and best line can be drawn through the extremes of the error bars to find the uncertainty in the gradient and intercept

1.4 Uncertainties

Types of Uncertainty

Absolute uncertainty: The uncertainty expressed in the same unit as the measurement, e.g., L = 25.0 ± 0.5 cm.
Percentage uncertainty: The uncertainty expressed as a percentage of the measured value. % uncertainty = (absolute uncertainty / measured value) × 100%.

Determining Uncertainty

- **From a single reading:** The uncertainty is at least ± the resolution of the instrument (e.g., ruler → ±0.5 mm for each end, so ±1 mm total for a length)
- **From repeat readings:** Uncertainty = range / 2 = (max value − min value) / 2. Use whichever gives the larger uncertainty.
- **Digital instruments:** Uncertainty = ± the last digit (e.g., 3.45 V → ±0.01 V)

Combining Uncertainties

Operation	Rule	Example
Addition or Subtraction (a = b + c or a = b − c)	Add absolute uncertainties $\Delta a = \Delta b + \Delta c$	If b = 5.0 ± 0.1 and c = 3.0 ± 0.2, then a = 8.0 ± 0.3
Multiplication or Division (a = b × c or a = b / c)	Add percentage uncertainties %a = %b + %c	If %b = 2% and %c = 3%, then %a = 5%
Raising to a power ($a = b^n$)	Multiply percentage uncertainty by the power %a = n × %b	If $a = b^3$ and %b = 2%, then %a = 6%

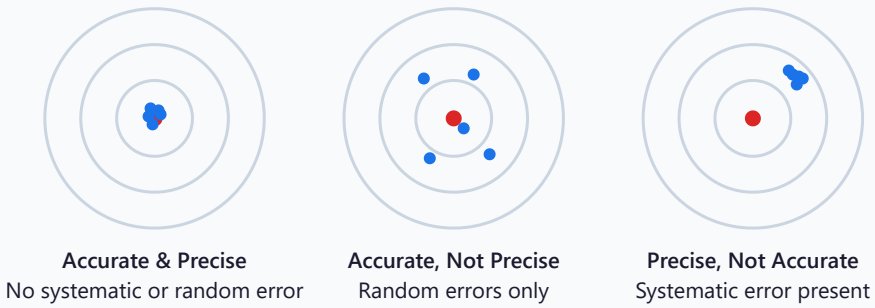
Exam Tip: When a quantity appears as a square (e.g., r^2 in $A = \pi r^2$), you must double its percentage uncertainty. Constants like π have zero uncertainty.

Uncertainty in Gradients and Intercepts

- Draw a **best-fit line** and a **worst-fit line** (the steepest or shallowest line that still passes through all the error bars)
- Uncertainty in gradient = |gradient of best-fit − gradient of worst-fit|
- Uncertainty in intercept = |intercept of best-fit − intercept of worst-fit|

1.5 Errors

Systematic vs Random Errors — Target Board Analogy



Systematic vs Random Errors — Target Board Analogy

Random Errors

Random errors: Unpredictable variations in measurements that cause readings to scatter about the true value. They affect precision.

- Caused by: fluctuations in readings, human reaction time, environmental disturbances
- Reduced by: taking repeat readings and calculating a mean, using data-logging equipment, plotting a graph and drawing a line of best fit
- Random errors cannot be eliminated entirely

Systematic Errors

Systematic errors: Errors that cause all readings to be shifted in the same direction (all too high or all too low) by a consistent amount. They affect accuracy.

- Caused by: zero errors on instruments, incorrectly calibrated equipment, parallax error (reading scales at an angle), background radiation not accounted for
- Reduced by: calibrating instruments, using appropriate techniques (e.g., reading at eye level), checking for zero errors before starting
- Systematic errors are not reduced by taking repeats

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Zero Errors

A zero error is a type of systematic error where an instrument does not read exactly zero when it should. Before taking measurements, check the zero reading and either adjust the instrument or subtract the zero error from all readings.

1.6 Accuracy and Precision

Accuracy: How close a measured value is to the true value. Affected by systematic errors.
Precision: How close repeated measurements are to each other. Affected by random errors. A precise set of results has a small spread.
Repeatability: The same person using the same equipment gets similar results.
Reproducibility: Different people using different equipment get similar results.

	High Accuracy	Low Accuracy
High Precision	Results clustered around the true value (ideal)	Results clustered together but away from the true value (systematic error)
Low Precision	Results scattered around the true value (random errors)	Results scattered and away from the true value (both error types)

1.7 Evaluating Experiments

Identifying Weaknesses

- Small range of IV → difficult to identify the relationship
- Insufficient repeat readings → cannot assess random error
- Low resolution instruments → high percentage uncertainty in small measurements
- Parallax error → systematic shift in readings
- Uncontrolled variables → unreliable conclusions
- Timing errors → human reaction time (~0.2 s) significant for short intervals

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Suggesting Improvements

- Use instruments with better resolution or precision (e.g., micrometer instead of ruler)
- Increase the number of repeat readings
- Use a fiducial marker to reduce parallax
- Use data-logging / sensors to remove human reaction time
- Time many oscillations and divide (e.g., time 20 swings, divide by 20) to reduce percentage uncertainty
- Use a larger range and more values of the IV
- Video recording with frame-by-frame analysis for timing
- Control temperature using a water bath or insulation

Exam Tip: In evaluation questions, always link the weakness to a specific improvement. Generic answers like "use better equipment" without explanation will not gain marks. State what the problem is, why it matters, and how the improvement addresses it.

Percentage Difference

To evaluate whether an experimental result agrees with a theoretical value:

$$\% \text{ difference} = |(\text{experimental} - \text{theoretical}) \,/\, \text{theoretical}| \times 100\%$$

If the percentage difference is within the percentage uncertainty of the experiment, the result is consistent with the theoretical value.

Worked Example: A student measures g = 9.52 m s^{-2} with an uncertainty of ±5%. The accepted value is 9.81 m s^{-2}. Does the result agree?

% difference = |9.52 − 9.81| / 9.81 × 100% = 0.29/9.81 × 100% = **3.0%**

Since 3.0% is within the experimental uncertainty of 5%, the result is **consistent** with the accepted value. If the percentage difference were larger than the uncertainty, there would be evidence of unaccounted systematic error.

Worked Example: Combining uncertainties. A cylinder has diameter d = 12.4 ± 0.1 mm and length L = 50.0 ± 0.5 mm. Calculate the volume and its percentage uncertainty.

$V = \pi d^2 L/4 = \pi(12.4 \times 10^{-3})^2(50.0 \times 10^{-3})/4 =$ **6.04×10^{-6} m³**

%unc in d = (0.1/12.4) × 100% = 0.81%

%unc in d^2 = 2 × 0.81% = 1.61% (power rule)

%unc in L = (0.5/50.0) × 100% = 1.00%

Total %unc in V = 1.61% + 1.00% = **2.6%**

Absolute uncertainty = $0.026 \times 6.04 \times 10^{-6} = 0.16 \times 10^{-6}$ m³

$V = (6.0 \pm 0.2) \times 10^{-6}$ m³